

Suggested problems: 1-6, 8, 9, 11

Problem 1: Suppose that a fair, six-sided die is rolled. Construct a probability measure P , including a description of the set of events.

Problem 2: Explain why $P(\emptyset) = 0$.

Problem 3: Explain why $P(A^c) = 1 - P(A)$, for all events A .

Problem 4: Explain why $P(A) \leq P(B)$, if A and B are events such that $A \subset B$.

Problem 5: Explain why $0 \leq P(A) \leq 1$ for all events A .

Problem 6: Explain why $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ for all events A and B .

Problem 7: A patient arrives at a doctor's office with a sore throat and low-grade fever. After an exam, the doctor decides that the patient has either a bacterial infection or a viral infection or both. The doctor decides that there is a probability of 0.7 that the patient has a bacterial infection and a probability of 0.4 that the person has a viral infection. What is the probability that the patient has both infections?

Problem 8: A contractor is building an office complex and needs to plan for water and electricity demand. After consulting with prospective tenants and examining historical data, the contractor decides that the demand for electricity will range somewhere between 1 and 150 (in millions of kilowatt-hours per day) and water demand will be between 4 and 200 (in thousands of gallons per day). All combinations of electrical and water demand are considered possible.

(a) Construct an appropriate sample space for all combinations of electrical and water demand.

(b) Construct an appropriate probability measure.

(c) The “high demand” event is when the water demand is at least 100 or the electricity demand is at least 115. What is the probability of this event?